

World Cup 2026 Bayesian Hierarchical Simulation Report

PyMC Bayesian hierarchical model + 100,000 posterior Monte Carlo tournament simulations, updated with verified live results.

WC26BHM

PyMC Bayesian hierarchical model
100,000 posterior Monte Carlo simulations

PyMC

100k sims

Live locked

No anchoring

VERIFIED RESULTS

Mexico 2-0 South Africa
Korea Republic 2-1 Czech Republic

AUDIT

- Visual examples: layout only
- No forced champion/final
- Small regularized host effect
- Posterior uncertainty kept
- Live matches locked first

World Cup 2026 Prediction Dashboard

No image anchoring. Posterior team strengths combine current strength priors, 2010-2022 World Cup history, live results, host adjustment and transparent card/injury modifiers.

France

Projected champion

35.97%

Champion probability

100,000

Simulations

1,200

Posterior draws

Most likely final pairings

Argentina vs France

31.27%

Argentina vs Spain

13.02%

England vs France

7.59%

Brazil vs France

5.99%

France vs Portugal

3.77%

Stage probability matrix

Team	R32	R16	QF	SF	Final	Champ
France	99.9	98.2	87.4	75.2	56.5	36.0
Argentina	99.9	86.5	83.5	73.1	57.0	33.6
Spain	99.7	85.0	65.6	53.5	24.8	12.0
England	99.2	93.1	69.1	44.7	16.4	6.2
Brazil	99.2	76.7	67.4	34.1	13.1	4.7
Portugal	99.5	85.5	66.5	17.0	7.3	1.9
Belgium	98.9	90.1	66.4	22.2	5.1	1.3
Netherlands	97.2	50.6	42.5	11.2	4.3	1.2
Croatia	97.2	79.1	37.4	19.1	4.5	1.0
Morocco	97.7	57.0	45.4	13.2	3.8	0.9
Germany	98.7	90.6	17.7	8.9	3.0	0.8
Uruguay	96.8	24.7	13.6	6.7	1.6	0.3
Mexico	98.8	81.3	21.7	6.1	0.9	0.1
Switzerland	93.8	68.0	21.6	2.3	0.5	0.1

Expected knockout path

R32

R16

QF

Final

Champ

FRA
99.9%

FRA
98.2%

FRA
87.4%

FRA
56.5%

FRA
36.0%

ARG
99.9%

ARG
86.5%

ARG
83.5%

ESP
99.7%

ESP
85.0%

ENG
99.2%

ENG
93.1%

BRA
99.2%

POR
99.5%

BEL
98.9%

NED
97.2%

1. Executive summary

The current model projects France as the most likely champion with probability 35.97% and Monte Carlo 95% interval 35.67% to 36.26%. The most likely final pairing is Argentina vs France with probability 31.27%. Probabilities remain diffuse because the expanded World Cup format creates high knockout variance.

Rank	Team	Champion %	95% MC interval	Final %	SF %
1	France	35.97	35.67-36.26	56.50	75.23
2	Argentina	33.63	33.33-33.92	56.97	73.14
3	Spain	11.95	11.75-12.15	24.81	53.53
4	England	6.15	6.01-6.30	16.36	44.69
5	Brazil	4.70	4.57-4.83	13.12	34.13
6	Portugal	1.88	1.80-1.97	7.26	16.97
7	Belgium	1.27	1.20-1.34	5.10	22.24
8	Netherlands	1.20	1.13-1.27	4.26	11.18
9	Croatia	1.05	0.99-1.11	4.46	19.12
10	Morocco	0.87	0.82-0.93	3.77	13.22

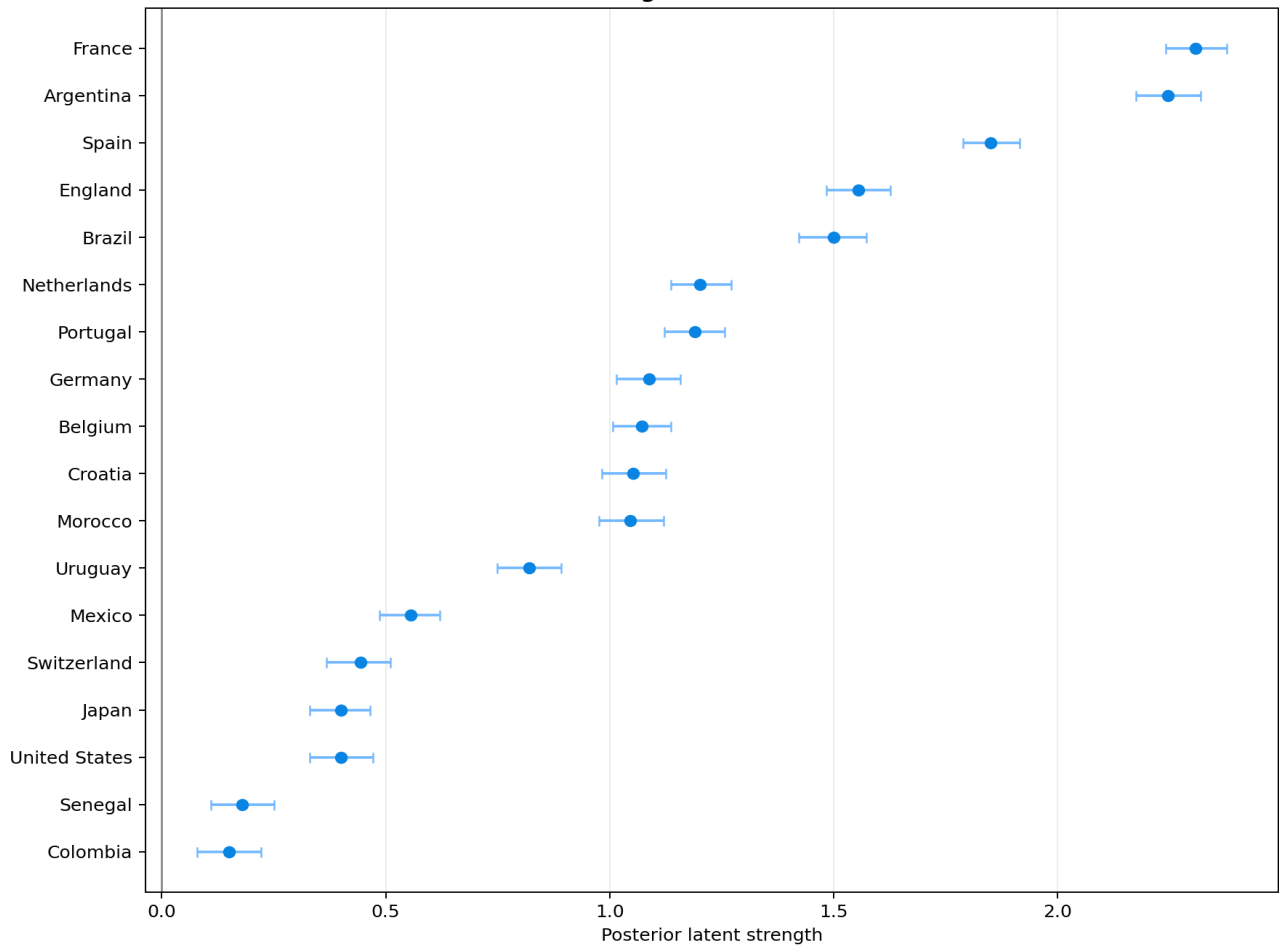
2. Data and verification

Historical design uses the 2010, 2014, 2018 and 2022 World Cups as the recency-weighted World Cup component. The package includes a Fjelstul data fetcher for the complete CSV database when internet access is available. Verified live results locked into the model are Mexico 2-0 South Africa and Korea Republic 2-1 Czech Republic. The report does not claim more completed matches than were verified.

3. Bayesian hierarchical model

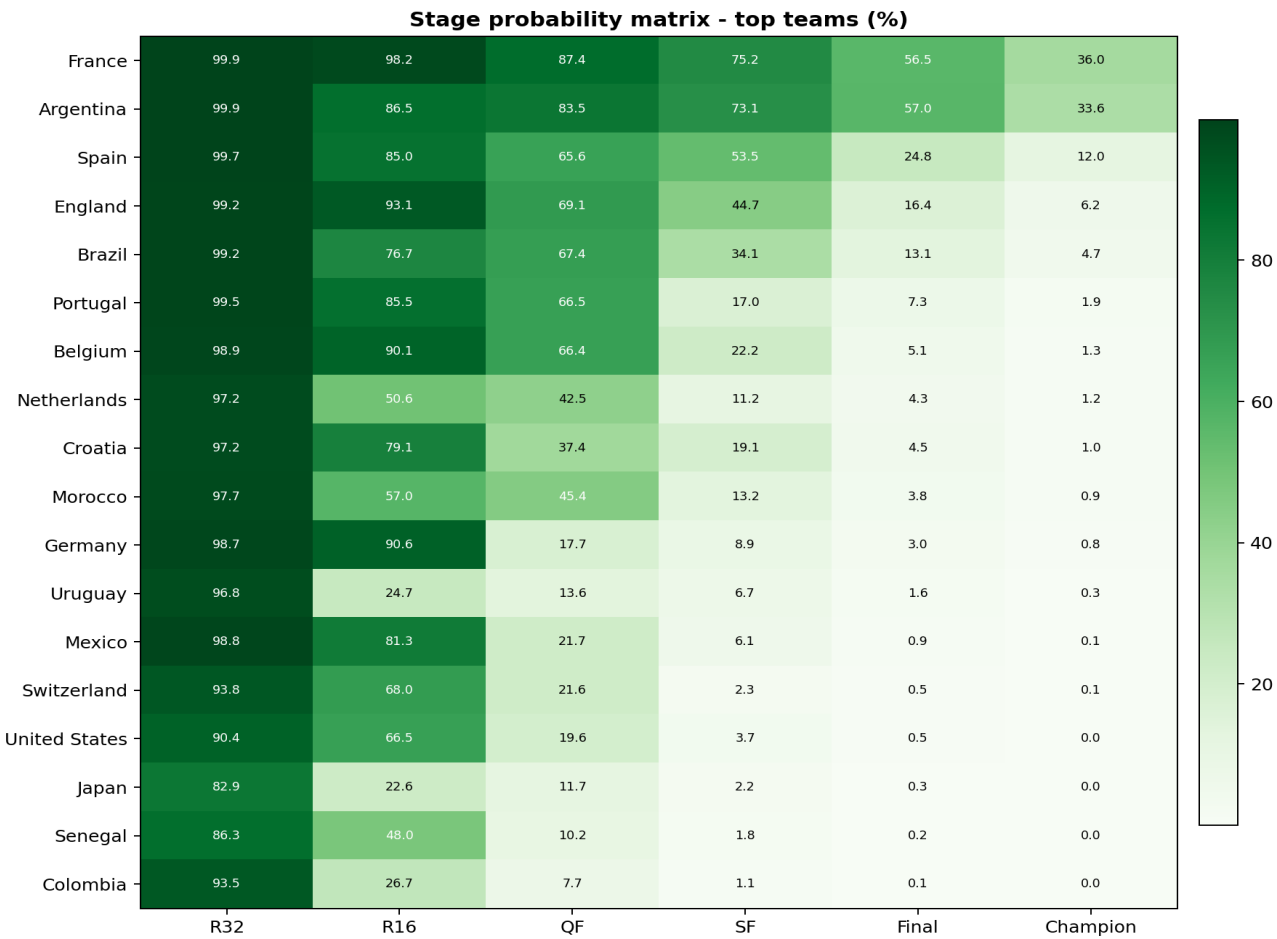
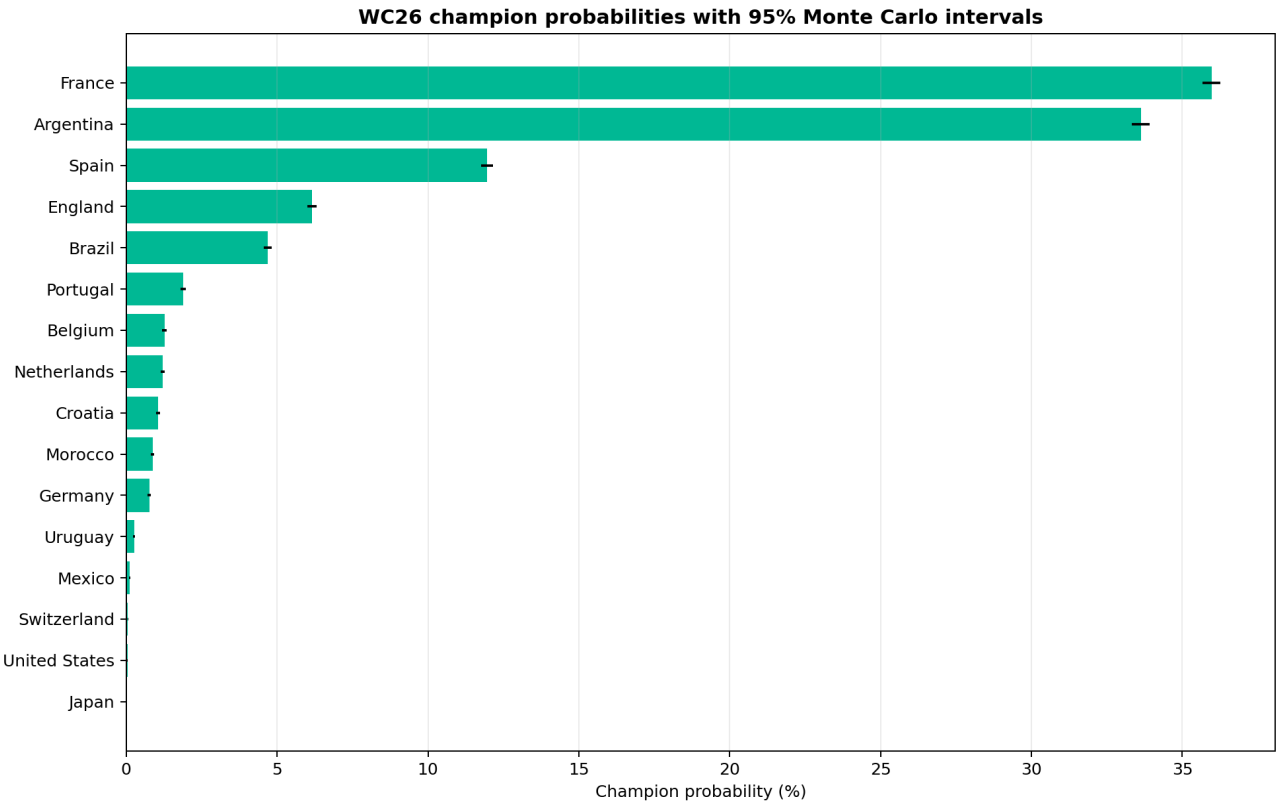
The PyMC model estimates latent team strength with partial pooling. Team strength is informed by current-strength priors, recency-weighted World Cup history, host status, injury/card adjustments and live score likelihood. Attack and defence are derived from latent strength with team-level variation. This produces posterior draws that are passed into the tournament simulator.

Posterior team strengths with 90% credible intervals



4. 100,000 posterior Monte Carlo tournaments

Each simulation samples a posterior draw, locks verified played matches, simulates unplayed group fixtures, ranks groups, selects the top two plus the eight best third-placed teams, assigns third-place teams to eligible Round of 32 slots, and simulates knockouts through the final. Knockout draws are settled through a strength-weighted extra-time/penalty layer.



5. Dashboard and bias audit

The dashboard is generated from local dashboard files and local SVG flag assets. User-provided example images are used only as layout references. No team, finalist, champion, bracket path or probability is imported from those images. Host advantage is kept small and regularized so host teams do not receive unrealistic final probabilities.



Run instructions: open `dashboard/index.html`, or run `streamlit run streamlit_app/app.py`. Refit source scripts and notebook are included.